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Paleomagnetism of Jurassic and Cretaceous rocks in central Patagonia: a key to constrain the timing of rotations during the breakup of southwestern Gondwana?

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Abstract

A paleomagnetic study in Jurassic and Cretaceous rocks from the Cañadón Asfalto basin, central Patagonia, indicates the occurrence of about 25–30° clockwise rotation in Upper Jurassic–lowermost Cretaceous rocks, whereas the overlying mid-Cretaceous rocks do not show evidence of rotation. This constrains the tectonic rotation to be related to a major regional unconformity in Patagonia, which in turn seems to be close in time with the early opening of the South Atlantic Ocean. The sense and probably the timing of this rotation are similar to those of other paleomagnetically detected rotations in different areas of southwestern Gondwana, suggesting a possible relationship between these and major tectonic processes related with fragmentation of the supercontinent. On the other hand, the mid-Cretaceous rocks in the region yield a paleopole located at Lat. 87° South, Long. 159° East, A95 = 3.8°. This pole position is consistent with coeval high-quality paleopoles of other plates when transferred to South American coordinates, implying it is an accurate determination of the Aptian (circa 116 Ma) geomagnetic field in South America. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Patagonia; tectonics; Mesozoic; Gondwana

1. Introduction

Patagonia was the locus of pervasive extensional tectonics during the early stages of Gondwana breakup [1–4]. Early to Middle Jurassic crustal stretching was mainly taken up by the development of NW–SE trending fault-bound basins (Fig. 1). This led to a regional basin and range depositional setting associated with widespread bimodal volcanism. By the Late Jurassic the areas affected by normal faulting enlarged and became interconnected at the time that the volcanic activity diminished. Most normal faulting ceased by the time that oceanic crust began to form in the southern South Atlantic (i.e. earliest Cretaceous).

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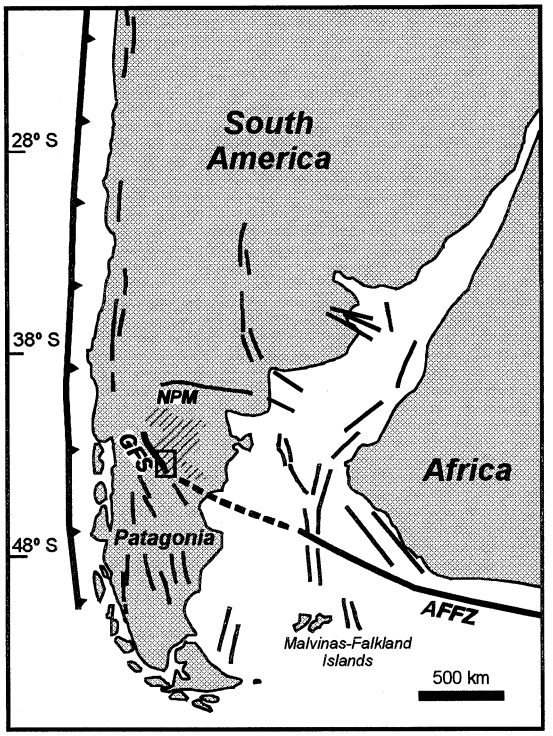


Fig. 1. Configuration of Western Gondwana previous to opening of the South Atlantic Ocean (South America in present coordinates) showing the main rift systems in the region (modified from Uliana et al. [4]). The location of the Agulhas–Falkland fracture zone (AFFZ), the Gastre Fault System (GFS) and the North Patagonian Massif (NPM) are shown. Dashed line is the proposed link between the GFS and the AFFZ throughout the Argentine continental shelf [19,20]. Hatched area in Patagonia depicts the Cañadón Asfalto Basin.

A regional unconformity between the Upper Jurassic–Neocomian and the mid-Cretaceous is a conspicuous feature throughout the region. Later, Cenozoic tectonic inversion locally affected rocks as young as Lower Tertiary.

Despite its history of crustal instability, paleomagnetic results from Jurassic and Cretaceous rocks in Patagonia have been used to construct apparent polar wander paths (APWP) for South America [5–7] without taking into account the possibility of vertical-axis rotations. In order to explore the effects of the Jurassic and younger tectonics on the paleomagnetic record, we performed paleomagnetic sampling in four areas in the Cañadón Asfalto basin, central Patagonia. This region is particularly pertinent for our purposes because its stratigraphy and structure have been recently refined by fieldwork carried out for oil exploration. Paleomagnetic results from Upper Jurassic–lower Neocomian rocks in one area indicate the occurrence of a clockwise tectonic rotation in a tens of kilometers-sized crustal block during the Early Cretaceous, whereas data from coeval rocks in other areas are inconclusive for tectonic analysis. This rotation is related to a major regional unconformity in Patagonia, and seems to be close in time with the early opening of the South Atlantic Ocean. Possible implications of these results in the framework of western Gondwana breakup are discussed.

2. Jurassic–Cretaceous evolution of the Cañadón Asfalto basin

The Cañadón Asfalto basin is located in the southern part of the North Patagonian Massif (Fig. 1). When describing its evolution, Figari et al. [8,9] divided the Mesozoic column into four major units or megasequences, each representing different stages in the development of the rift basin (Fig. 2). We used this megasequence classification as the stratigraphic guide to carry out the paleomagnetic work.

The evolution of the Cañadón Asfalto basin started during the Late Triassic–Early Jurassic with the development of NW–SE-oriented halfgrabens with their depocenters to the SW. The early depositional record comprises conglomerates and sandstones grouped into the Megasequence 0 (MS0). During the Middle Jurassic to early Late Jurassic, the halfgrabens were filled by a thick volcaniclastic unit (Lonco Trapial Formation), followed by black shales and calcareous lake deposits with interbedded basaltic flows (Lower Cañadón Asfalto Formation). Both these unit are grouped into the Megasequence 1 (MS1).

Fault reactivation during latest Jurassic created room for two depositional cycles which constitute the Upper Jurassic to Neocomian Upper Cañadón Asfalto Formation or Megasequence 2

(MS2). The lower cycle comprises lacustrine mudstones and limestones, whereas fluviodeltaic sandstones, siltstones, and mudstones compose the upper cycle. A new reactivation tilted the rocks of MS1 and MS2 producing the regional unconformity at the base of Megasequence 3 (MS3), which is composed of local fluvial strata (Los Adobes Formation) and a widespread pyroclastic unit with conspicuous development of paleosols (Cerro Barcino Formation). Paleontological data and paleomagnetic results ([6] and this paper) constrain the sampled rocks in MS3 to be Upper Barremian?–Aptian.

The basin underwent tectonic inversion during the Cenozoic leading to outcrops of depocenters along the active borders of the halfgrabens. However, the extensional structures are preserved away from these tectonically inverted zones.

3. Paleomagnetism

3.1. Sampling

Two hundred thirty six oriented core and block samples of volcanic and sedimentary rocks were taken from ten localities, representing four areas sparsely distributed over a region of more than 5000 km^2 (Fig. 2). Sampled areas were selected in order to obtain information about the regional unconformity that separates MS1–MS2 from MS3. Thus, paleomagnetic data from at least one locality corresponding to either MS1 or MS2, and another locality representing MS3, are available for each area. All sections in sedimentary rocks involved stratigraphic packages from about 40 to 600 m thick, from which several fine-grained beds were sampled. Typically one to five samples were taken from each bed, and at least 15 samples from each section. This procedure was followed in order to average out the paleosecular variation (PSV) within the locality. A similar procedure was followed by sampling several flows in volcanic localities. Additional data of the region that we used in this study are those reported by Somoza [6] for rocks of MS3.

Most Jurassic–Neocomian rocks (MS1 and MS2) are variably tilted throughout the region. In contrast, most sampled mid-Cretaceous rocks (MS3) are subhorizontal, being tilted only where Cenozoic tectonic inversion occurred.

3.1.1. Area 1: Sierra de Pichiñanes (Fig. 2b)

All samples of this area belong to basaltic flows of the Lower Cañadón Asfalto Formation (MS1). The outcrops are continuous on both sides of the Chubut River (Fig. 2c), dipping 25° towards the southwest. A 200 m thick section was sampled in two locations (CA and BA sections, 54 samples) along the west side of the river. Samples in the northern location (BA) were obtained above and below a 10 m thick limestone bed which is intercalated in the middle part of the basaltic section. Only the lower levels of the section were sampled in the southern location (CA). An additional location (PCH section, 10 samples) was sampled in a 30 m thick section located on the eastern side of the river. The MS3 is represented by the LP localities from Los Adobes and Cerro Barcino Formations whose results have been previously reported by Somoza [6]. These mid-Cretaceous rocks crop out in the eastern side of the Sierra de Lonco Trapial (Fig. 2), being variably tilted during Cenozoic tectonic inversion. Farther to the east they are subhorizontal, overlying the basaltic outcrops in the western margin of the Chubut River (Fig. 2a).

3.1.2. Area 2: Gorro Frigio depocenter (Fig. 2b)

Outcrops of the Upper Cañadón Asfalto Formation (MS2) are particularly widespread in this area (Fig. 2a). We took samples (CX location, 48 samples) in a thick section of this unit, where strata dip about 25° to the northwest. Thirty-nine mudstones to fine-grained sandstones samples correspond to the upper sedimentary cycle, whereas the remaining nine samples are mudstones of the lower cycle. Farther to the northwest we took 28 samples (K location, Fig. 2a) in another section of the upper sedimentary cycle where strata dip 7° towards the northeast. A few kilometers eastward of the CX section, we sampled the MS3 (LC section, 18 samples) where Los Adobes and Cerro Barcino Formations lie subhorizontal.

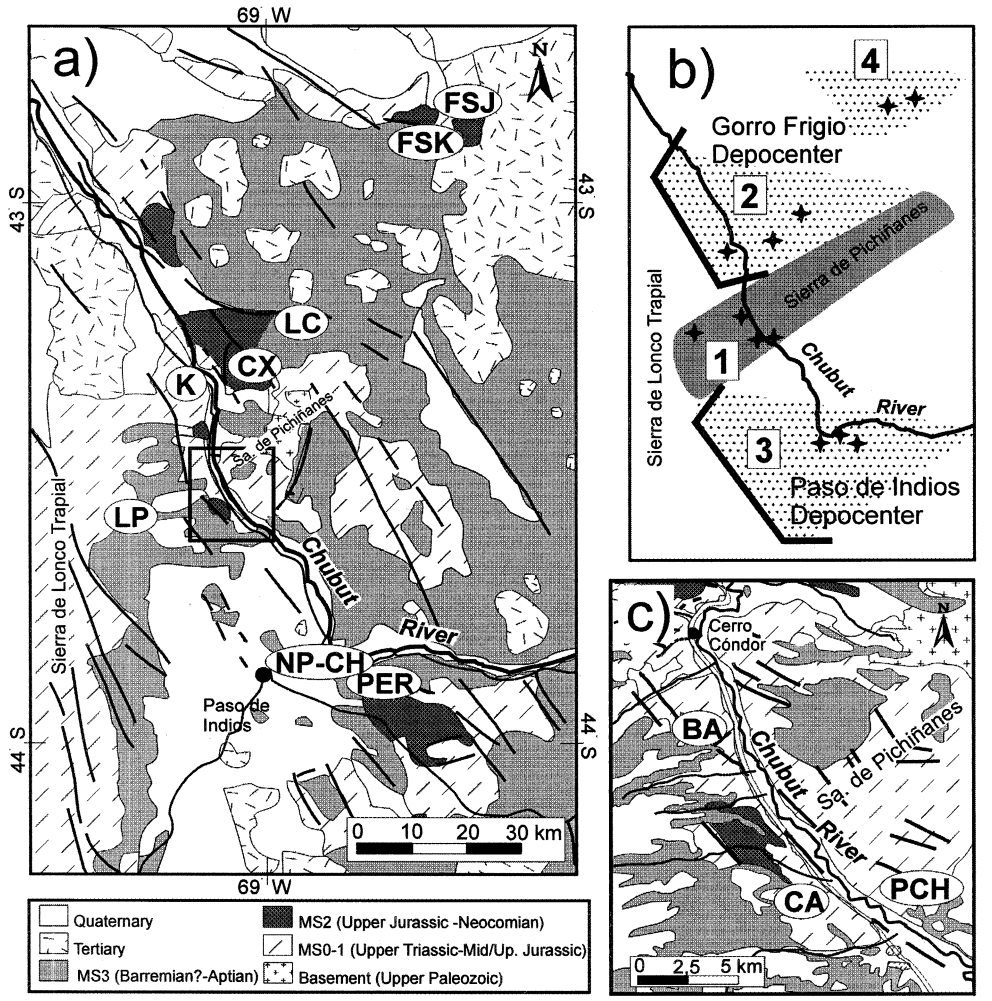


Fig. 2. (a) Geologic map of the studied region showing the location of the paleomagnetic sections. The window indicates the sampling area of the basalts from MS1 (c). (b) Main morphotectonic features in the region, with numbers showing the four areas with paleomagnetic data: area 1 is Sierra de Pichiñanes; area 2 is Gorro Frigio depocenter; area 3 is Perdomo at Paso de Indios depocenter; and area 4 is Estancia Fossatti. (c) Detailed map of the sampling area of the basalts from MS1 showing location of the sections.

3.1.3. Area 3: Perdomo (Fig. 2b)

In this area there are isolated outcrops of the Upper Cañadón Asfalto Formation (MS2). The more extensive is near the Chubut River elbow, where rocks dipping 20° to the northwest crop out below subhorizontal layers of the MS3. Due to the poor exposure of the MS2, the only way that we could sample a relatively thick section of this unit was taking samples following a continuous topographic level a few meters (~4) below the basal conglomerate of the MS3. We obtained 22 samples from several stratigraphic levels in a 100 m thick section (PER location, Fig. 2a), but this sampling procedure had important implications for our results (see Section 3.3.3 of this paper). The MS3 in this area is represented by the CH location (Cerro Barcino Formation) reported by Somoza [6]. However, new stratigraphic work in this area (E. Figari et al., unpublished mapping, 1997) indicates that some subhorizontal outcrops previously believed to be mid-Cretaceous actually correspond to the Upper Cañadón

Asfalto Formation. This is the case for fluvial strata that include CH2 and CH3 sites in Somoza [6]. Accordingly, these sites were removed from the MS3 data, and five additional beds (NP section, 25 samples) of the Cerro Barcino Formation were sampled a few kilometers to the west where mid-Cretaceous strata dip gently (2–3°) towards the SW (Fig. 2a).

3.1.4. Area 4: Estancia Fossatti (Fig. 2a)

In this area we took mudstone samples (FSJ location, 15 samples) from a 150 m thick section of the lower part of the Upper Cañadón Asfalto Formation (MS2). The rocks dip 13° towards the northeast, and are exposed in the hanging wall block of a Cenozoic reverse fault. In the foot wall block of the same fault, a few hundred meters from the cliff that marks the fault trace, we took 16 samples (FSK location) in a 40 m thick package of redbeds of the Los Adobes Formation (MS3) where strata dip 12° towards the south.

3.2. Methods

At least two standard specimens of each sample were subjected to stepwise demagnetization. Remanence measurements were done with either a three-axis 2G Enterprises cryogenic magnetometer or a modified DIGICO spinner magnetometer. Alternating field demagnetization used either a 2G600 or a prototype tumbling demagnetizer. Thermal demagnetization was carried out employing a Schonstedt TSD-1 furnace. Bulk magnetic susceptibility was measured after each thermal step in order to monitor possible magnetic mineral changes. Isothermal remanent magnetization (IRM) was applied to identify the minerals that carry the magnetization.

Paleomagnetic components observed in the specimens were determined using principal component analysis [10]. A magnetic component was accepted only if the maximum angular deviation (MAD) was less than 10°. A mean direction for each stratigraphic level was obtained by averaging magnetic components from specimens of each sedimentary bed or volcanic flow. In some cases this procedure was performed by combining magnetic components with demagnetization circles [11]. The obtained single-stratigraphic-level directions within each section were averaged using standard Fisher [12] statistics to obtain section-mean directions.

Time-averaged virtual geomagnetic poles (VGP) derived from section-mean directions were compared with reference paleopoles for stable South America. As most Mesozoic South American APWP available in the literature contain paleopoles from Patagonia, we constructed an alternative set of reference poles by combining paleomagnetic data of South America (excluding Patagonia) and other plates. The criteria followed for paleopole selection as well as the reconstruction parameters used to transfer the data to South American coordinates are described in the Appendix.

3.3. Description and interpretation of the results

In this part we describe the paleomagnetic behavior of the samples and discuss the origin of the observed paleomagnetic directions. First we present and discuss the data from the MS3 in the entire region. Then, data of the MS2 and MS1 from each particular area are presented.

3.3.1. Los Adobes and Cerro Barcino Formations (MS3)

The intensity of the natural remanent magnetization (NRM) ranged from 0.1 to 20 mA/m. Paleomagnetic behavior from samples of these mid-Cretaceous units was straightforward. A single component of normal polarity, which decays towards the origin, was isolated in all the specimens (e.g. Fig. 3a). Paleomagnetic behavior and rock magnetic analysis indicate that this component is carried by magnetite in samples from the Cerro Barcino Formation at the localities LC and NP (areas 2 and 3, respectively), and a mixture of magnetite and hematite in samples from Los Adobes Formation in section LC (area 2) (Fig. 3b). This is in agreement with previous results from other localities involving the same units in areas 1 and 3 [6]. In contrast, IRM experiments indicate that hematite is the only magnetic mineral present in samples from the Los Adobes Formation at section FSK, area 4 (Fig. 3b). Petro-

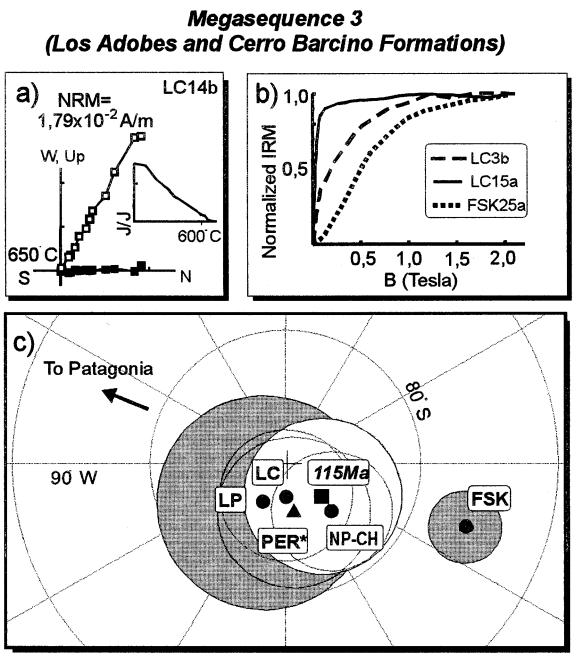


Fig. 3. Paleomagnetic results from rocks of the Los Adobes and Cerro Barcino Formations (MS3). (a) Example of typical demagnetization behavior of rocks of MS3, illustrated by a specimen of the Cerro Barcino Formation at location LC. Open (solid) squares indicate projection onto the vertical (horizontal) plane. (b) Typical examples of acquisition of IRM for samples of the Cerro Barcino Formation (specimen LC15a from section LC), and Los Adobes Formation (specimen LC3b from section LC, and specimen FSK25a from section FSK). Note that nearly absence of magnetite distinguishes the rocks from the FSK locality (dotted curve) from those of the other locations of MS3. (c) The time-averaged VGPs from the locations LP, LC, and NP-CH are concordant with the 115 Ma reference pole reported in the Appendix (square). The VGP derived from the tilt-corrected direction of locality FSK is discordant with the remaining VGPs of MS3. We suggest this location was remagnetized during or after tilting in the Cenozoic (see discussion in Section 3.3.1). The VGP derived from the in situ mean-direction of Upper Jurassic–Neocomian rocks from locality PER (triangle) is concordant with the VGPs derived from rocks of MS3 in the region, suggesting the rocks of MS2 at location PER underwent remagnetization during the mid-Cretaceous (see discussion in Section 3.3.3).

graphic studies on rocks of the Los Adobes Formation from localities LP and LC showed a primary sedimentary fabric where magnetite is present as poorly to moderately rounded grains, whereas hematite occurs as fine-grained pigment within the matrix and thin oxidation rims in some grains. In contrast, rocks from section FSK showed a pervasive growth of diagenetic hematite, which disrupted the primary sedimentary fabric. These optical observations correlate with the differences showed by the rock magnetic results.

Mean directions from rocks of MS3 in each area are reported in Table 1. The ratio of the precision parameters for the LP sites ($K_{\text{untilted}}/k_{\text{in situ}} = 14.66$, $n = 5$, [6]) indicates that the magnetization carried by these rocks at area 1 is pretectonic at the 99% confidence level [13]. MS3 rocks from areas 1, 2, and 3 show identical paleomagnetic behavior, the same carriers of magnetization, and locality-mean directions indistinguishable from one to another, suggesting a common origin for their magnetizations. The exclusive presence of normal polarities suggests these magnetizations were acquired during the Cretaceous Normal Superchron. This is consistent with Aptian fossil assemblages founded in the Cerro Barcino Formation at Perdomo [14]. However, we can not exclude the possible presence of upper Barremian strata in the Los Adobes Formation since the possibility of sampling the earliest Aptian M0 chron is low due to its relatively brief duration.

In contrast, the tilt-corrected mean direction from the FSK location (area 4) is removed from the group obtained from the other three locality-means, the difference residing in its lower inclination (see its derived VGP in Fig. 3c). As mentioned above, the only magnetic mineral detected in these rocks is secondary hematite. This, together with the proximity of this section to the fault trace (see Section 3.1.4) and the fact that its in situ section-mean direction is similar to that of the Cenozoic magnetic field (see Table 1), leads us to suggest a remagnetization linked to tectonic inversion in the Cenozoic. Then, the results of the FSK locality will be ignored for tectonic analysis.

The VGPs derived from MS3 in areas 1, 2, and 3 are concordant when compared with the mid-Cretaceous reference pole reported in the Appendix (Table 2 and Fig. 3c). This suggests that this region of Patagonia has not undergone significant vertical-axis deformation since Late Barremian?–

Aptian time. We calculated a mid-Cretaceous (circa 116 Ma) paleopole for South America by combining VGPs derived from paleomagnetic directions of localities LC (two directions), CH (four directions), NP (five directions), and LP (five directions): Lat. 87° South, Long. 159° East, A95 = 3.8°, $K = 100$, $N = 16$. This pole position is removed only 2.4° from the paleopole reported by Somoza [6] which it supersedes, and is also indistinguishable from the alternative paleopole obtained by combining the three time-averaged VGPs from the areas 1, 2, and 3: Lat. 87° South, Long. 168° East, A95 = 4°, $N = 3$ localities.

3.3.2. Upper Cañadón Asfalto Formation (MS2) in the area 2 (Gorro Frigio depocenter)

NRM intensity ranged from 0.1 to 10 mA/m. Most specimens from the lower sedimentary cycle (lower part of CX section) showed just a soft magnetization which was eliminated at 30 mT or 400°C, and were rejected for further analysis. In contrast, a component of magnetization showing a high unblocking temperature and high coercivity was isolated after removing a softer component in 57% of the demagnetized specimens from the 67 samples of the upper sedimentary cycle (K and upper part of CX sections, Fig.

Table 1
Paleomagnetic data for the Cañadón Asfalto Basin

Site	*N*/*n*	Mean direction (in situ)				Tilt strike/dip	Mean direction (tilt-corrected)					VGP	
		Dec.	Inc.	α_{95}	*k*		Dec.	Inc.	α_{95}	*k*	Lat.	Long.	dp/dm
Area 1: Sierra de Pichiñanes													
MS3: Cerro Barcino and Los Adobes Formations													
LP	5/25	38.5	−48.3	22.8	12.0	[a]	4.5	−62.5	5.8	176.0	−86.7	208.5	7.0/9.0
MS1: Lower Cañadón Asfalto Formation (volcanics)													
BA2	3/7	164.6	66.2	14.2	76.1	133/25	192.3	47.7	14.2	76.1	−72.3	148.7	12.1/18.5
BA1+CA1+PCH3	11/49	66.2	65.5	3.8	144.3	[b]	131.7	73.5	4.1	123.8	−57.0	335.2	6.6/7.4
PCH2	1/3	154.7	67.2	–	–	136/27	191.2	49.9	–	–	−74.3	149.3	–
PCH1	1/4	248.9	−67.2	–	–	136/27	349.7	−79.6	–	–	−63.3	298.8	–
Area 2: Gorro Frigio Depocenter													
MS3: Cerro Barcino and Los Adobes Formations													
LC-CB	5/9	0.0	−58.8	3.4	492.5	Subhoriz.	0.0	−58.8	3.4	492.5	−86.3	111.2	3.8/5.1
LC-LA	3/9	8.3	−64.7	6.6	346.0	Subhoriz.	8.3	−64.7	6.6	346.0	−83.1	234.1	8.6/10.7
LC Mean	8/18	2.8	−61.1	3.5	248.4		2.8	−61.1	3.5	248.4	−87.6	178.8	4.1/5.4
MS2: Upper Cañadón Asfalto Formation													
CX	17/35	180.3	47.5	6.7	29.1	222/25	211.7	58.8	6.7	29.1	−66.7	207.6	7.4/10.0
K	8/19	202.8	49.3	9.7	33.2	300/5	201.9	54.2	9.7	33.2	−71.4	184.1	9.6/13.7
CX and K mean	25/54	187.3	48.5	5.8	25.9		208.3	57.4	5.4	30.0	−68.5	201.2	5.7/7.8
Area 3: Perdomo-Paso de Indios Depocenter													
MS3: Cerro Barcino Formation													
NP	5/24	358.8	−61.3	4.6	270.0	127/3	1.3	−59.4	4.6	270.0	−86.0	127.0	5.0/7.0
CH	4/27	3.8	−58.4	8.3	120.0	Subhoriz.	3.8	−58.4	8.3	120.0	−84.0	144.0	9.0/12.0
NP and CH mean	9/51	1.1	−60.0	3.8	180.0		2.4	−58.9	3.6	200.0	−85.0	136.0	4.0/5.0
MS2: Upper Cañadón Asfalto Formation													
PER in situ	22/22	4.2	−60.8	4.2	54.0						−86.3	169.9	5.0/6.5
PER						221/20	44.6	−66.9	4.2	54.0	−59.0	230.4	5.8/7.0
Area 4: Estancia Fossatti													
MS3: Los Adobes Formation													
FSK	7/16	356.6	−60.6	2.5	581.2	90/12	359.5	−48.7	2.5	581.2	−75.9	109.8	2.2/3.3
MS2: Upper Cañadón Asfalto Formation													
FSJ	4/4	186.2	54.8	25.4	14.0	38/13	173.6	46.7	25.4	14.0	–	–	–

N/*n*: Number of beds (flows)/number of samples used in statistics. α_{95}, *k*: Fisher [12] statistical parameters. LP and CH sites after Somoza [6].

[a]For values of tectonic correction see [6].

[b]BA1 133/25; CA1 104/24; PCH3 136/27.

Table 2
Rotations computed in the Cañadón Asfalto Basin

Site (Loc.)	Age	Ref.	R	ΔR	P	ΔP
MS3						
LP (1)	Barremian (Barr.)–Aptian (Apt)	115	+2.9	10.8	−3.7	7.9
LC (2)	Barr.–Apt	115	+1.6	8.0	−2.4	6.1
CH (3)	Apt	115	+2.8	12.5	+1.2	9.6
NP (3)	Apt	115	−0.6	8.8	+0.1	6.7
MS2						
K (2)	Upper Jurassic (UJ)–Neocomian (Neoc)	130	+24.9	11.5	+3.9	9.4
CX (2)	UJ–Neoc	130	+34.6	9.2	−0.8	7.1
K+CX (2)	UJ–Neoc	130	+31.3	7.1	+0.7	5.6
[K+CX] rev	UJ–Neoc	130	+33.1	7.6	−3.9	6.2
K (2)	UJ–Neoc	145	+20.5	12.9	+15.7	10.1
CX (2)	UJ–Neoc	145	+29.6	10.9	+10.9	8.0
[K+CX] (2)	UJ–Neoc	145	+26.4	9.2	+12.5	6.7
[K+CX] rev	UJ–Neoc	145	+28.6	9.6	+15.7	7.2

Ref.: Reference pole used to calculate rotations (see Appendix and Table 5)
R, ΔR: rotation (clockwise is positive) and P, ΔP: poleward apparent translation (northward transport is positive), calculated in VGP space by the method of Beck et al. [36].

4a,b). Acquisition of IRM suggests that the carriers of this component are magnetite and hematite in varying proportions. Usually both minerals are present in the same specimen. The high temperature component could not be successfully isolated in 21% of the collection, with a persistent overlap of two components. For these specimens, demagnetization circles were used to determine the characteristic direction. The remaining specimens carried just a soft component and were discarded.

Six polarity zones were detected in the upper part of the CX section, and three polarity zones in the K section. Both of these two locations yield a positive, class C, reversal test [15] (Fig. 4c). A tilt test [16] performed between the normal directions is indeterminate due to the low number of normal directions available in the section K, whereas the result between the reversed directions is positive. The tilt test involving the whole data from the CX and K sections is also positive at the 95% confidence level (Fig. 4c). Then, both of these two localities show similar mean directions when referred to the paleohorizontal, suggesting the southern part of the Gorro Frigio depocenter acted as a single structural block with respect to vertical-axis rotational deformation since Early Cretaceous.

A mean direction for MS2 at area 2 is given in Table 1. Regardless of whether the entire data set from sections CX and K or the reversed directions only are considered, the resulting VGP is discordant with respect to the reference paleopoles reported in the Appendix (Table 2 and Fig. 4d), indicating a clockwise rotation of about 25–30°. We estimated the map-view area of the rotated crustal block to be about 1000 km^2 (Fig. 2). Additionally, the age of this rotation is constrained to be Early Cretaceous since paleomagnetic data from the overlying MS3 at nearby section LC are concordant with its corresponding reference direction (Fig. 3c).

3.3.3. *Upper Cañadón Asfalto Formation (MS2) in the area 3 (Perdomo)*

AF demagnetization was not useful in isolating a magnetic component in the sampled rocks of the Upper Cañadón Asfalto Formation (MS2) at Perdomo. Thermal demagnetization, however, allowed isolating a high temperature component carried by a high coercivity magnetic mineral (Fig. 5b). In contrast with the common presence of reversed polarities in rocks from this unit in other areas, only normal polarities were observed at Perdomo (Fig. 5a). The geologic significance of this magnetization is not obvious. The untilted section-mean direction shows a clockwise declination anomaly (Fig. 5a), which is consistent with

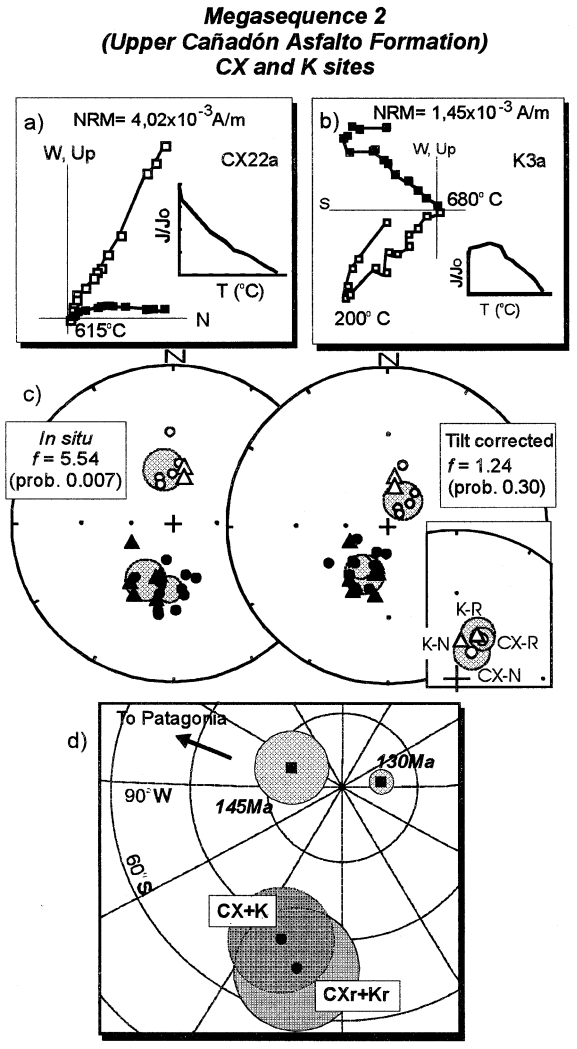


Fig. 4. Results from rocks of the Upper Cañadón Asfalto Formation (MS2) at Gorro Frigio depocenter (area 2). (a,b) Examples of a typical demagnetization behavior illustrated by a specimen from section CX (a), and a specimen from section K (b). Conventions as in Fig. 3. (c) Paleomagnetic directions obtained from different stratigraphic levels in locality CX (circles) and K (triangles). The cones of 95% confidence levels for the normal and reversed mean-directions at each section are shown (mean-direction for two strata with normal polarity magnetization at the site K has no α_{95}). Bedding correction produces an improvement in the grouping of the mean site directions; statistical values of the test using the whole data set (25 sites) yield $f = 5.54$ at 0 percent unfolding, and $f = 1.24$ at 100 percent unfolding, which indicates a positive fold test [16]. The same is observed when the test is applied on 18 reversed directions only ($f = 4.36$ vs. 0.73 in situ and unfolded respectively). This strongly suggests that the magnetization was acquired pretilting. Inset depicts a positive reversal test for this paleomagnetic data (reversed directions inverted into normal directions). (d) The resulting time-averaged VGP after combining all the data for localities CX+K is indistinguishable from the VGP obtained using reversed polarity directions only (CXr+Kr). The CX+K VGP is discordant with respect to both the 130 and 145 Ma reference poles reported in the Appendix, indicating the Gorro Frigio depocenter (Fig. 2) experienced a clockwise tectonic rotation (Table 2). The age of these rocks (uppermost Jurassic–Neocomian) together with the results from section LC in this area (Fig. 3) indicate that tectonic rotation occurred during the Early Cretaceous and before Late Barremian?–Aptian times.
←

the presence of a tectonic rotation. However, the in situ mean direction is concordant with the paleomagnetic results from rocks of MS3 (Fig. 3c showing its corresponding VGP), suggesting that the rocks of MS2 in this locality underwent remagnetization during the mid-Cretaceous. We favor the later interpretation on the basis of the exclusive presence of normal polarities (Fig. 5a), the different magnetic carrier (Fig. 5b), and the sampling procedure we followed for MS2 at Perdomo (see Section 3.1.3). In this case we virtually sampled the bedrock of the basal conglomerate in the overlying Cerro Barcino Formation. Then, we suggest that mid-Cretaceous weathering transformed the primary magnetic phases of the sampled strata into fine-grained hematite, recording the Aptian geomagnetic field.

3.3.4. Upper Cañadón Asfalto Formation (MS2) in the area 4 (Estancia Fossatti)

NRM intensity ranged from 0.1 to 1 mA/m. As mentioned above, only the lower cycle of the Upper Cañadón Asfalto Formation was sampled in this area. Most specimens showed behavior similar to that observed in samples from the lower cycle at locality CX, and were rejected. Stable components were isolated for two stratigraphic levels, and demagnetization circles were determined in specimens of another two beds. Available data for this section belong to only four samples (Table 1), which makes the results useless for tectonic analysis.

3.3.5. Lower Cañadón Asfalto Formation (MS1) in the area 1 (Pichiñanes)

NRM intensities of these rocks, which range from 100 to 3000 mA/m, correlate with their degree of deuteric oxidation. The least oxidized black lavas showed the highest NRM intensities and susceptibility, the lower coercivity spectrum, and the steeper IRM acquisition curves. On the other hand, higher oxidation in the reddish-brown lavas resulted in lower NRM intensities, higher coercivity spectrum, and IRM acquisition at higher magnetizing fields. Despite the different magnetic carriers, all the specimens carried a highly stable component isolated up to 70 mT or 650°C in black and reddish lavas, respectively (Fig. 6a,b).

Directions from the CA location and the lower part of the BA location are indistinguishable from one another, suggesting these flows were erupted in a short time-interval. Likewise, all the samples from flows in the upper part of the BA location show similar paleomagnetic directions, suggesting that another short-duration volcanic event is recorded above the 10 m thick limestone bed. On the other hand, directions from samples of the PCH locality suggest the presence of three different eruptive events in the eastern margin of the Chubut River (Fig. 6d).

We conclude that only a few time recordings of the paleomagnetic field are provided by the 230 m thick basaltic section in the area 1. Moreover, some of these paleomagnetic directions show very high inclination with a strong declination departure (e.g. CA section and lower part of BA section), suggesting insufficient sampling of PSV. Accordingly, the results from the MS1 in the area 1 are inconclusive for tectonic analysis.

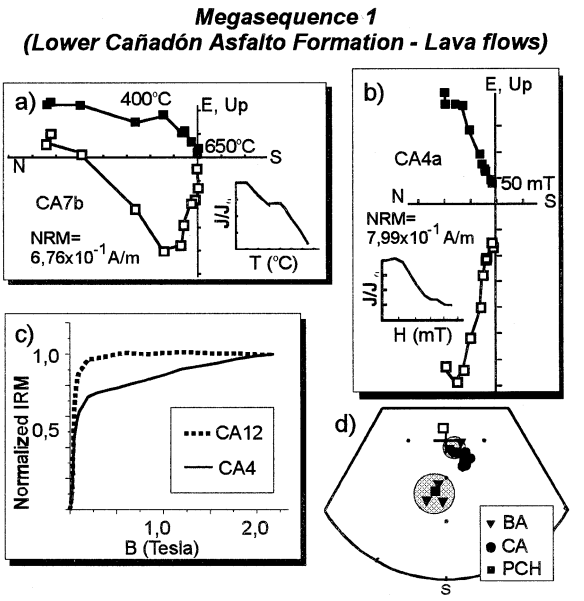


Fig. 6. Paleomagnetic results from basalts of the Lower Cañadón Asfalto Formation (MS1) at the Sierra de Pichiñanes (area 1 in Fig. 2). Conventions in demagnetization diagrams as in Fig. 3. (a) Result after stepwise thermal demagnetization on a sample from location CA. (b) Result after stepwise AF demagnetization on a sample from location CA. (c) IRM acquisition in samples from location CA. (d) Tilt-corrected directions from flows of locations BA, CA, and PCH. Note that only a few, directionally distinct records of the ancient magnetic field seem to be present in this 200 m thick basaltic section.

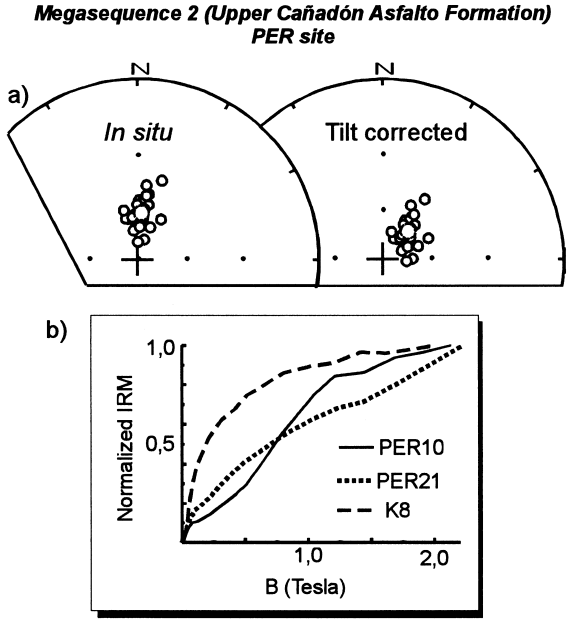


Fig. 5. Results from rocks of the Upper Cañadón Asfalto Formation (MS2) at Perdomo (area 3). (a) Contrasting with results from this unit in other localities, all the samples from MS2 at Perdomo carry normal polarity magnetizations. (b) IRM results indicate that most (if not all) the magnetic minerals in these rocks are (fine-grained?) hematite (dotted and solid curves), whereas a mixture of magnetite and hematite carry the magnetization of samples from this unit at other localities (dashed curve).

3.3.6. Summary

Several of our results are not suitable for tectonic analysis. Paleomagnetic data from several flows in a more than 200 thick basaltic section of MS1 at area 1 failed in averaging out PSV. Paleomagnetic directions obtained from MS2 at area 3 suggest the occurrence of a remagnetization associated with mid-Cretaceous weathering. Results from the area 4 have either poor resolution (MS2 in FSJ section) or evidences of remagnetization (MS3 in FSK section).

In contrast, the results from MS3 in areas 1, 2, and 3 suggest that the studied region has acted as a coherent structural block with respect to vertical-axis deformation since Aptian times. Moreover, the time-averaged paleomagnetic pole from these rocks of MS3 is concordant with coeval paleopoles of other plates transferred to South American coordinates, suggesting structural integrity of the region with respect to cratonic South America since mid-Cretaceous (Fig. 3c).

On the other hand, results from Upper Jurassic–Neocomian rocks at area 2 indicate that an area of nearly 1000 km^2 underwent a clockwise tectonic rotation of about 25° (Fig. 4d). Results from MS3 in this area constrain the age of this tectonic rotation to be Early Cretaceous.

4. Discussion

During latest Jurassic to earliest Cretaceous, Patagonia was a piece of Gondwana subjected to widespread extension bordered by the evolving Southern Atlantic rift system to the east, and a zone of active convergence to the west (Fig. 1). Our results indicate that Upper Jurassic to lowermost Cretaceous rocks of the Cañadón Asfalto basin underwent clockwise vertical-axis rotation prior to deposition of overlying Upper Barremian? to Aptian rocks. This rotation seems to be related to extensional deformation in central Patagonia. Although available data do not permit an estimate of the extent of the zone affected by rotational deformation, several lines of evidence suggest it could be greater than the area sampled in this study.

An important tectonic feature in central Patagonia is the poorly understood Gastre Fault System which was originally described as a NW–SE left lateral strike–slip system throughout the studied region [17] (Fig. 1). On the other hand, Rapela and Pankhurst [18] proposed that this fault system acted as a dextral shear zone during the Early to Middle Jurassic. The rotation we document could be kinematically consistent with this last proposal, but its timing is different. At a greater regional scale, recent interpretations suggest that this fault system is part of a major Jurassic to Early Cretaceous intracontinental dextral-shear zone that also involved the Agulhas–Falklands Fault zone in the southernmost Atlantic (Fig. 1) [18–21]. However, both the large amounts of strike–slip motion attributed to the Gastre Fault System in some of these models as well as its proposed eastward continuation across the Argentine continental shelf are difficult to reconcile with the regional geology of Patagonia (e.g. [22]). Despite uncertainties about the regional extent and timing of displacement on the Gastre Fault System, it could have been an important structure to accommodate extension (or transtension) and associated rotations in a wide region of Patagonia during the early stages of western Gondwana breakup.

In recent years, the important role of rotations (from small crustal blocks to microplates) in accommodating deformation during the breakup of southwestern Gondwana has been recognized [21,23–28], although differences in tectonic interpretations indicate a complex history of deformation. For instance, paleomagnetic results from Jurassic dolerite dykes in the western Falkland (Malvinas) Islands (Fig. 1) suggest the occurrence of a large clockwise rotation. Both the mechanism and the timing of this rotation are disputed (e.g. [21,25,28]). In particular, Thomson [28] argues for a Valanginian age of rotation, which implies similarity in time with rotation of the Cañadón Asfalto Basin. At a greater regional scale, Early Cretaceous clockwise rotations older than 110 Ma also have been found in several microplates of Western Antarctica [24,26,29].

Although the consistency in sense and timing of these tectonic rotations is suggestive, we accept that any possible relationship remains speculative.

Nevertheless, our results suggest that further paleomagnetic work on Jurassic and Cretaceous rocks in Patagonia could help to better understand the kinematics and timing of deformations during the early stages of western Gondwana breakup. On the other hand, our paleomagnetic results suggest that rotational deformation in central Patagonia ceased by the time sea floor spreading started in the Southern Atlantic. As a consequence, paleomagnetic studies on mid-Cretaceous and younger rocks from Patagonia could yield a suitable record of the ancient South American geomagnetic field, whereas data from older rocks should be treated with caution for APWP purposes.

5. Conclusions

Our paleomagnetic results in central Patagonia provide evidence for about 25–30° clockwise rotation in a tens of kilometers sized crustal block during the Early Cretaceous. Although we can not resolve unambiguously the relationship with tectonic rotations detected in other regions of southwestern Gondwanaland, the available paleomagnetic data seem to suggest that clockwise rotations occurred in widely separated areas during the Early Cretaceous. This encourages to perform further systematic paleomagnetic and isotopic studies of Jurassic and Cretaceous rocks in Patagonia. Potential results could provide a better understanding of deformation during the breakup of Gondwana.

On the other hand, our paleomagnetic data indicate that mid-Cretaceous and younger rocks from Patagonia are useful for determination of the APWP of South America. We obtained an updated Aptian (circa 116 Ma) paleopole located at Lat. 87° South, Long. 159° East, A95 = 3.8°.

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Appendix

Late Jurassic–mid-Cretaceous reference poles for South America

Although the ages of the sampled rocks in this study span from Middle Jurassic to mid-Cretaceous, the older ones (MS1) did not give suitable information for tectonic analysis (see 3.3.5). Thus, for our purposes we determined only reference poles spanning from Late Jurassic to mid-Cretaceous. To do this, we compiled paleomagnetic data from different continents and determined three reference poles with mean-ages at 145 ± 5, 130 ± 5, and 115 ± 5 Ma. The two older paleopoles are useful for their comparison with data from MS2, whereas the younger paleopole is appropriate for analyzing MS3 results. The time-scale used is that of Gradstein et al. [30].

Methods and results

Requisites for paleopole selection and acceptance were: (1) well defined magnetizations and evidence supporting successful averaging of the PSV; (2) the age of the magnetization to be acceptably constrained; (3) no unresolved structural complications and reasonable structural coherence of sampled region with the craton. Rather than a large data set including paleopoles of lesser quality, we preferred working with a small number of high quality poles derived from multiple plates, whenever possible. This procedure was useful to diminish any possible bias produced by a particular plate-reconstruction, although it could not be applied for all the time-intervals in this study. Table 3 lists the selected paleopoles.

Many finite rotation parameters between plates have been refined in recent years by better satellite-derived gravity field and additional magnetic data. We used some of these updated reconstruc-

Table 3
List of the selected poles

Mean age	Paleopole			Transferred paleopole		Rock unit	Ref.
	λ	ϕ	A95	λ	ϕ		
North American poles							
115	−73.5	12.9	3.9	−85.5	66.7	New England plutons	[a]
147	−67.5	341.8	4.0	−81.7	291.3	Upper Morrison Formation	[37]
147	−68.3	336.2	4.8	−80.1	280.9	Brushy Basin Member	[38]
147	−65.3	340.9	3.5	−80.2	301.6	Trujillo Hill	[39]
Eurasian poles							
140	−75.0	359.5	3.2	−88.6	268.0	Berriasian stratotype	[40]
South American poles							
125	−86.0	75.9	3.3			Sierra de los Cóndores Group	[41]
129	−85.3	69.3	2.9			Alkaline Province Paraguay	[42]
132	−84.1	64.4	2.3			Central Paraná basalts	[42]
132	−83.0	71.4	2.4			Northern Paraná basalts	[42]
133	−84.0	106.2	1.5			Southern Paraná basalts	[42]
Australian poles							
112	−49.0	149.0	4.0	−85.2	156.0	Otway group	[43]
Indian poles							
117	−8.0	117.0	3.0	−84.6	156.5	Rajmahal Traps	[44,45]
African poles							
132	−48.3	86.6	3.2	−84.7	103.9	Kaoko basalts	[46]

[a]See Appendix.

tion parameters to transfer the selected poles to South American coordinates (Table 4). Rather than interpolating, we used the published reconstruction nearest in age to the paleopole to be transferred. Some additional considerations about the choice of reconstructions are mentioned below. We consider that the obtained reference poles (Table 5) satisfactorily represent the ancient magnetic field in South America.

Further comments

A paleomagnetic pole calculated using VGPs

Table 4
List of the rotations used in transferring paleomagnetic data from one plate to another

Anomaly	Age	Lat.	Long.	Angle	Ref.
Eurasia to North America					
Closure	92.0	−69.34	−33.2	23.61	[47]
North America to Africa					
M0	120.5	66.41	−19.61	54.12	[47]
M17	141.5	67.07	−18.05	59.23	[47]
Africa to South America					
M0	120.5	51.60	−35.00	−52.90	[35]
Closure	~138.0	45.50	−32.20	−57.50	[34]
Australia to East Antarctica					
Closure	~95.0	−3.43	46.39	−28.27	[48]
India to East Antarctica					
M0	120.5	−0.80	9.70	−81.97	[49]
East Antarctica to Africa					
	110.0	−6.30	−31.80	33.16	[49]
M0	120.5	−7.00	−30.60	38.36	[49]

Clockwise rotation is negative.

Table 5
Combined mean paleopoles in South American coordinates

Age	Age span	Lat.	Long.	A95	*N*
115	110–120	−86.3	132.3	5.9	3
130	125–135	−84.8	81.8	1.7	6
145	140–150	−82.8	290.1	4.9	4

from five 122–119 Ma plutons in northeastern North America (Mount Ascutney [31], Mount Pawtuckaway [31], Alfred Complex [32], Tatnic Complex [32], and Cape Neddick Complex [32]; combined paleopole located at Lat. 72.6 N, Long. 195.3 E, A95 = 6.3°, $N = 5$) is indistinguishable from a younger paleopole derived from three 113–111 Ma intrusions in this region [33]. Combining these 8 VGPs we obtained a Late Barremian?–Aptian pole position for North American plate (Table 2). The earliest Aptian M0 anomaly (circa 120 Ma) is well constrained in the Atlantic and Indian oceans, accordingly its derived reconstructions were used to transfer most mid-Cretaceous poles. However, an available 110 Ma reconstruction for the Africa–Antarctica pair was applied to the paleopole of the 112 Ma Otway Group (Australia). The Cretaceous poles yielded a tight cluster when transferred to South American coordinates suggesting a proper determination of a synthetic Aptian pole position.

South America has several high-quality Early Cretaceous (circa 130 Ma) pole positions, and no larger data set is necessary. However we added a further high quality paleopole from the Kaoko lavas in Namibia (Table 3) which are the African counterpart of the southern Paraná basalts (Table 3). To transfer the Kaoko paleopole we used a rigid-plate reconstruction [34], since it produces better fit between these two unambiguously coeval poles than a modern reconstruction involving internal deformation in western Gondwana [35] (0.7 vs. 4.9°).

No paleopoles of latest Jurassic–earliest Cretaceous age from the Gondwana landmasses passed our selection. However, three North American and one European poles were selected for this time span. To transfer these four paleopoles to South American coordinates we used classical (i.e. rigid-plate) reconstructions. The use of models with possible differential motion between northwest and south Africa during the Mesozoic do not change our interpretation and, on the other hand, there are uncertainties in the amount and age of the intraplate motions in Africa.

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